

Class - 11

Types of events

Probability

INDEPENDENT EVENT

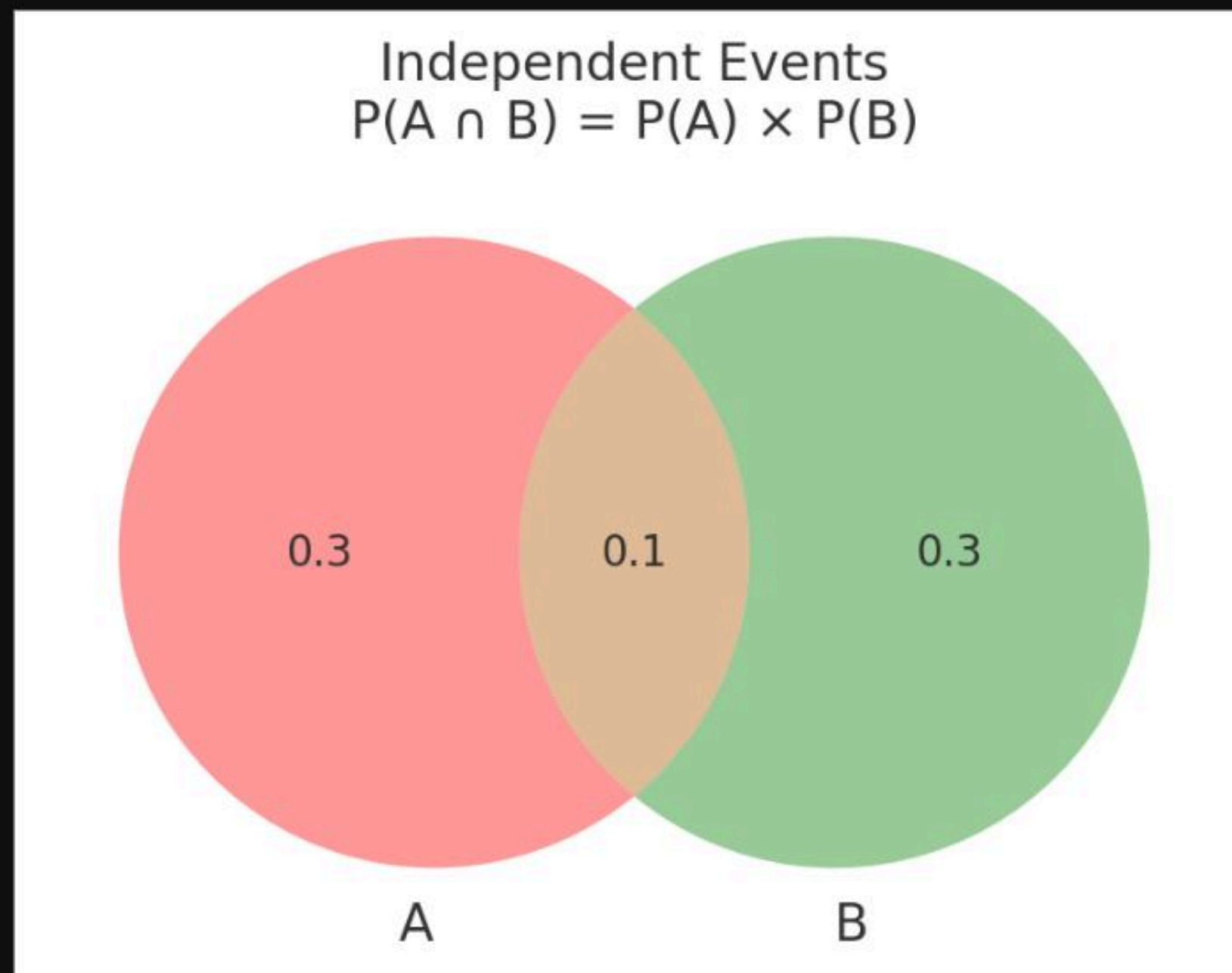
Two event A and B are said to be independent if the occurrence of one does not effect the probability of other.

If the probability of A does not change by the occurrence of probability of B is known as independent event.

$$P(A \cap B) = P(A) \cdot P(B)$$

This means that the probability of both events happening together is simply the product of their individual probability

INDEPENDENT EVENT



Venn Diagram

INDEPENDENT EVENT



Example:

- Event A: Roll a dice $\rightarrow$ Get a 6.
- Event B: Flip a coin $\rightarrow$ Get heads.

These are **independent** because the dice result doesn't affect the coin flip.

$$P(A) = \text{probability of rolling a 6 on a fair die} = \frac{1}{6}$$

$$P(B) = \text{probability of getting Heads on a fair coin} = \frac{1}{2}$$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

INDEPENDENT EVENT

Let's take another example:

- Event **A**: "Pick a **red card** from a standard 52-card deck."
- Event **B**: "**Light bulb turns on** when you flip the switch."

- $P(A) = \frac{26}{52} = \frac{1}{2}$

- $P(B) = \frac{90}{100}$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \times \frac{90}{100} = \frac{9}{20}$$

DEPENDENT EVENT

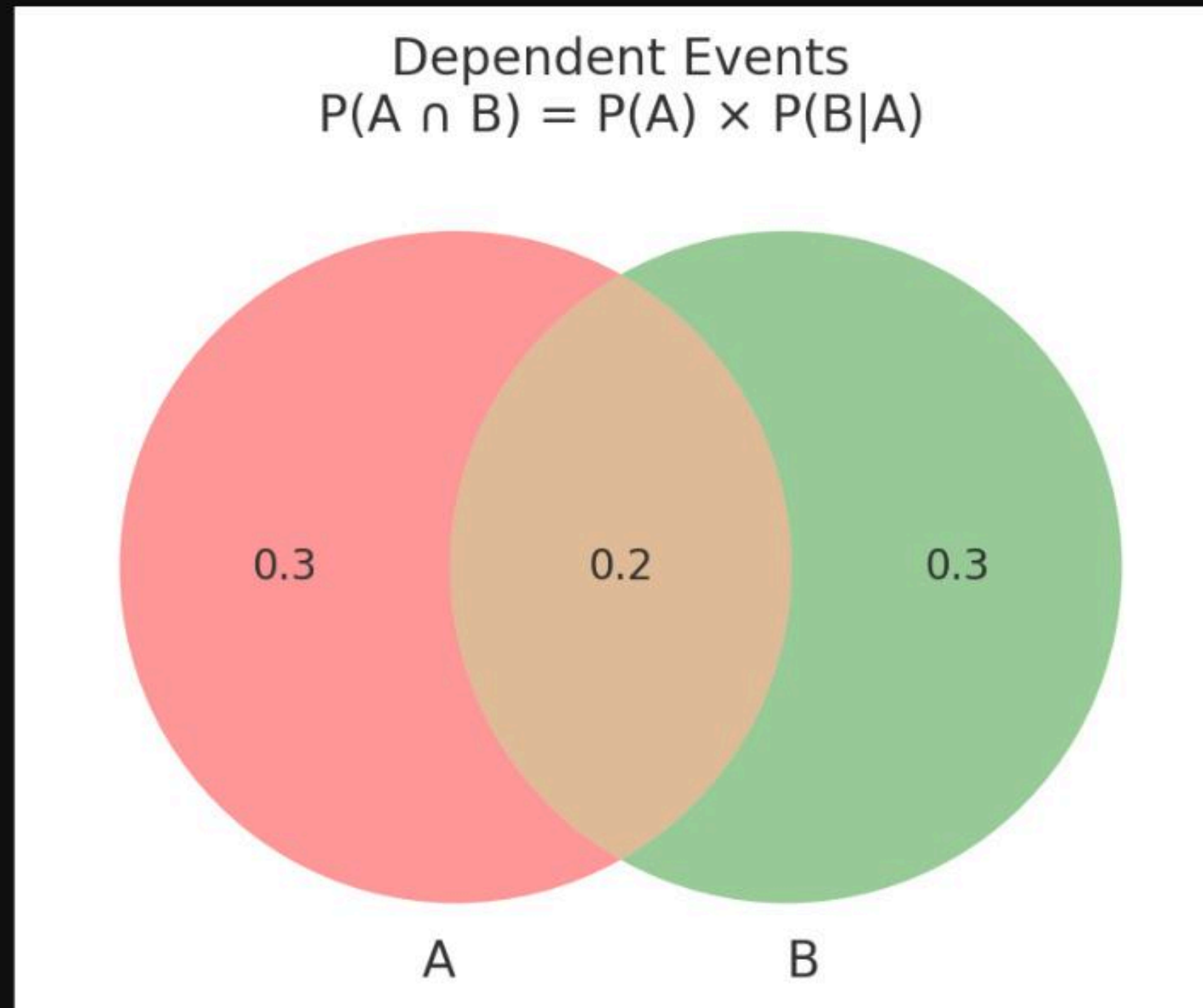
A **dependent event** is when the outcome of one event **affects** the probability of the other.

By dependent event probability we understand that by happening of $P(A)$ there is a change in the $P(B)$.

$$P(A \cap B) \neq P(A) \cdot P(B)$$

$$P(A \cap B) = P(A) \times P(B/A)$$

DEPENDENT EVENT



Venn Diagram

DEPENDENT EVENT

Example (cards without replacement):

- Event A = drawing an **Ace** first.
- Event B = drawing another **Ace** second.

- $P(A) = \frac{4}{52}$

- $P(B/A) = \frac{3}{51}$

- $P(A \cap B) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$

DEPENDENT EVENT

Let's take another Example: Bag of Balls ⑧ (without replacement)

- A bag has **3 red balls** and **2 blue balls** (total = 5).
- You pick **one ball, don't put it back**, then pick another.
- **Event A:** First ball is red.
- **Event B:** Second ball is red (after first was already red).

- $P(A) = \frac{3}{5}$

- $P(B|A) = \frac{2}{4} = \frac{1}{2}$

$$P(A \cap B) = P(A) \times P(B/A) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$$

MUTUALLY EXCLUSIVE EVENT

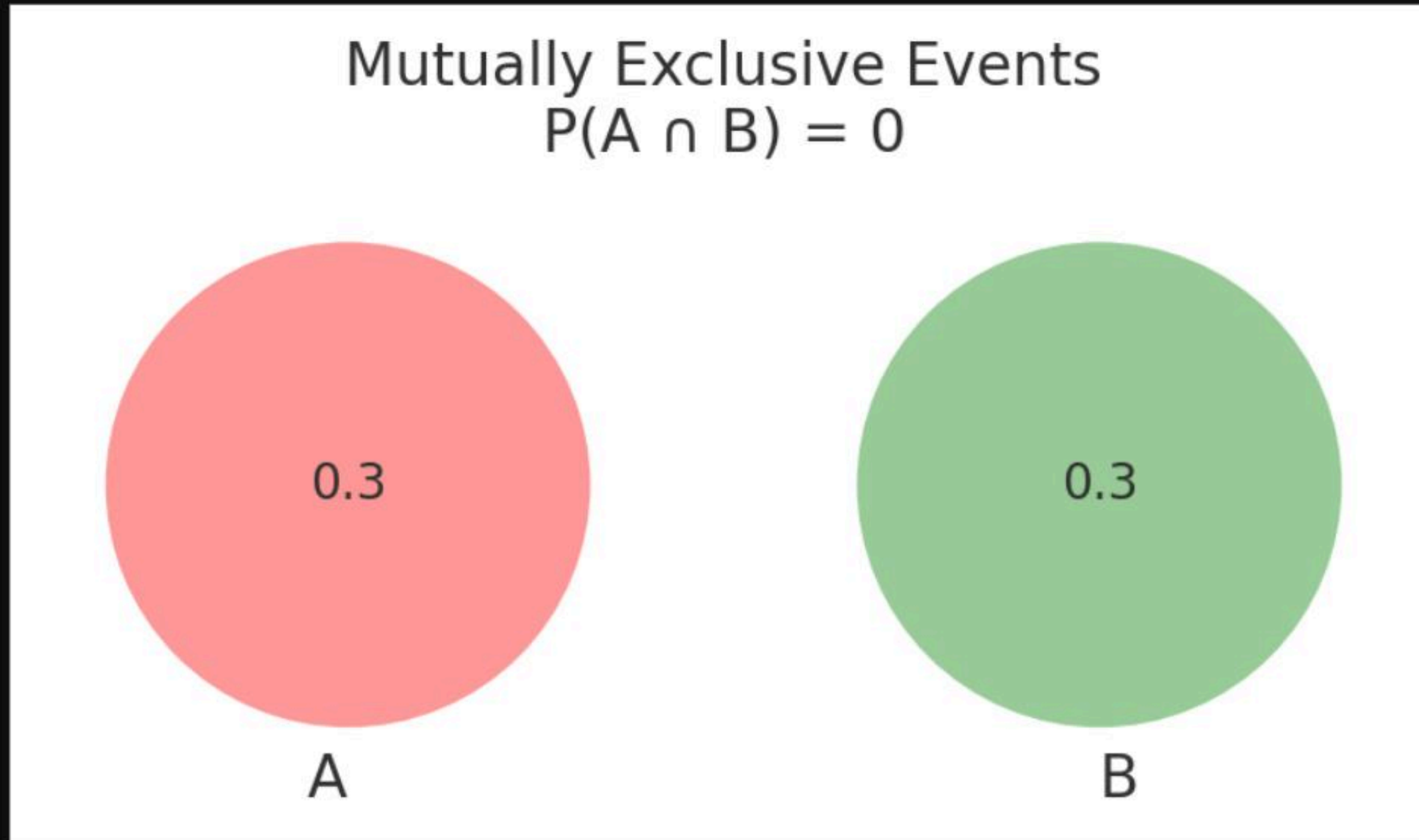
Two events are **mutually exclusive** if they **cannot occur at the same time** during a single trial of an experiment.

Mutually exclusive event are those events which can not be happen together at the **same time**

$$P(A \cap B) = 0$$

This means the joint probability of A and B is **zero**

MUTUALLY EXCLUSIVE EVENT



Venn Diagram

MUTUALLY EXCLUSIVE EVENT

Example of **Mutually Exclusive Event** 🎲 :

- **Event A:** Roll a die and get a **3**.
- **Event B:** Roll the same die and get a **6**.

$$P(A \cap B) = 0$$

Because in a single throw we can either get a 3 or a 6 in the dice but we don't get the both at the same time in a dice